

Draft version

Dynamical systems time series classification

Kirill Semkin^{1*} and Vadim Strijov^{2*}

^{1*}Moscow Russia.

^{2*}Moscow Russia.

*Corresponding author(s). E-mail(s): semkin.ki32@gmail.com;
strijov@forecsys.ru;

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1 Introduction

The paper introduces a method for time series classification. The *time series* is a finite sequence of scalar values $x(t_i) = \{x_{t_1}, \dots, x_{t_n}\}$ with variable length defined at particular times $t_1, \dots, t_n \in \mathbb{R}$. The *multivariate* time series is a sequence of vector values $\mathbf{x}(t_i) = \{\mathbf{x}_{t_1}, \dots, \mathbf{x}_{t_n}\}$ where $\mathbb{R}^d$. This is equal to having d time series with the same times. Everywhere further we assume time series to be multivariate in general. From theoretical point of view it is convenient to assume time series values $\mathbf{x}_{t_i}$ to be *random vectors*. Time series have an associated *class label* $y \in 1 \dots K$ that is also random in general. Here K is the number of classes. A *classifier* is a map $\delta(\mathbf{x}(t_i))$ that

associates any time series with a class label. We will seek to obtain a "good" classifier in some sense defined later.

There are many methods for time series classification in the literature. *Distance-based* approaches rely on measuring similarity between time series using distance metrics. The metric can be defined using Dynamic Time Warping [1] and the k-Nearest-Neighbors (k-NN) classifier can be used then [2]. *Feature-based* approaches extract finite set of statistical [3, 4] or spectral [5] features from time series for further classification. *Deep learning* approaches uses neural networks to automatically learn time series features from raw train data. Convolutional neural networks [6, 7], recurrent neural networks (RNN) [8] or their combination [9–11] are designed to process sequential data like time series.

In the paper we assume particular model behind time series - *dynamical system*. By that we mean the time-invariant *ordinary differential equation (ODE)* of the form

$$\begin{cases} \frac{d}{dt}\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}}), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_0, \end{cases} \quad (1)$$

where $\bar{\mathbf{x}} \in M$, $f(\mathbf{x}) : M \rightarrow M$ is a *vector field*, $\bar{\mathbf{x}}_0$ is a *starting point*, M is a differentiable manifold in general. We consider vector fields are not arbitrary but belong to some normed space $f \in (\mathcal{F}(M), \|\cdot\|)$. To find a solution $\bar{\mathbf{x}}(t)$ satisfying (1) on some time interval $t \in [0, T]$ is to solve an *initial value problem (IVP)*. We also denote IVP as $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)$ and call it *phase trajectory*. The IVP is also equivalent to the following integral equation

$$\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f) = \bar{\mathbf{x}}_0 + \int_0^t f(\bar{\mathbf{x}}(s|\bar{\mathbf{x}}_0, f)) ds. \quad (2)$$

Under some regularity conditions on f the solution is unique [12]. We associate K different vector fields f_j , $j \in 1, \dots, K$ with each series class. The series values are the

noisy phase trajectories

$$\mathbf{x}(t_i) = \bar{\mathbf{x}}(t_i) + \boldsymbol{\epsilon}(t_i), \quad (3)$$

where $\boldsymbol{\epsilon}(t_i)$ is a random noise process. Its values $\boldsymbol{\epsilon}_{t_i}$ are i.i.d.random vectors (mutually for all classes and any times) having

$$\mathbb{E}[\boldsymbol{\epsilon}_{t_i}] = 0, \text{ Cov}(\boldsymbol{\epsilon}_{t_i}) = R \text{ where } \|R\|_2 \leq \sigma^2 < \infty. \quad (4)$$

Here R is covariance matrix, $\|R\|_2$ is matrix spectral norm. Therefore $\mathbb{E}[\mathbf{x}(t_i)] = \bar{\mathbf{x}}(t_i)$ and $\text{Cov}[\mathbf{x}(t_i)] = R$. In general, the classifier must account for both the deterministic component $\bar{\mathbf{x}}(t_i)$ and the noise component $\boldsymbol{\epsilon}(t_i)$ of the series. We will later show that under some conditions there is an asymptotically accurate classifier that would not rely on the noise. Then we will discard these condition and will introduce more general classifier. Moreover, we will be able to adapt the classifier to the more general dynamical system described by the *stochastic differential equation (SDE)* [?] of the form

$$\begin{cases} d\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}})dt + GdW(t), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_0, \end{cases} \quad (5)$$

where $W(t)$ is a m -dimensional Brownian motion, G is a $d \times m$ constant diffusion matrix, unique for each class. Phase trajectories $\bar{\mathbf{x}}(t)$ are *random processes* in this model. The definition of the IVP translates straightforwardly for the SDE. The IVP solution is unique under some regularity conditions on f [?].

Our approach can be extended for the less direct connection between observed time series and the phase trajectories (3). Imagine the connection is given by some *observation map* $\alpha : M \rightarrow \mathbb{R}^d$ such that

$$\mathbf{x}(t_i) = \alpha(\bar{\mathbf{x}}(t_i)) + \boldsymbol{\epsilon}(t_i), \quad (6)$$

where M has dimension $k \neq d$ in general. We would like to reconstruct the $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$ to build a classifier. The reconstruction will be based on building an *embedding* of the manifold M . In general, an embedding is a smooth map $\phi : M \rightarrow N$ into another manifold N that is a diffeomorphism onto its image $\phi(M)$. It follows that phase trajectories $\bar{\mathbf{x}}(t) \subset M$ and $\phi(\bar{\mathbf{x}}(t)) \subset \phi(M)$ are basically identical (more strictly, diffeomorphic). To reconstruct the phase trajectories we need to build $\phi(\bar{\mathbf{x}}(t_i))$ from the observed time series $\mathbf{x}(t_i)$. One way to archive this is the Whitney Embedding Theorem [13]. If α is a generic map [?] from M to $\mathbb{R}^{2k+1}$ then $\alpha = \phi$ and therefore $\mathbf{x}(t_i) = \phi(\bar{\mathbf{x}}(t_i))$. This is basically the original connection (3) but with the diffeomorphic phase trajectories instead of the original one. However, this approach requires an excessive $2k + 1$ number of time series even when k is rather small. Another approach is the Takens embedding theorem [14]. It requires only one observed scalar time series $x(t_i)$. If α is a generic map from M to $\mathbb{R}$ then the embedding is given by the map

$$\phi(\bar{\mathbf{x}}) = (\alpha(x), f(\alpha(x)), \dots, f^m(\alpha(x))),$$

where $m > 2k$. Substituting $\bar{\mathbf{x}}(t_i)$ gives us $\phi(\bar{\mathbf{x}}(t_i)) = (x(t_i), x(t_{i-1}), \dots, x(t_{i-k}))$ which we call *delay vector*. Illustration of the Takens theorem is on the Fig. 1. The delay vectors form noisy diffeomorphic phase trajectory because $x(t_i)$ is itself noisy. So they are basically identical to $\mathbf{x}(t_i)$ from (3). Further, we would not differentiate reconstructed phase trajectories and the original one.

Our time series' model can be applied to many real-word classification problems. For example, in medicine an electroencephalogram (EEG) is a direct, macroscopic measure of the brain's electrical activity [15]. The activity is governed by the Maxwell's Equations that can be reduced to the ODEs [16]. The EEG can be used for detecting arrhythmias or other heart conditions [6] as well as to identify seizures and sleep stages from brainwave data [7]. In the weather forecasting the fundamental equations of

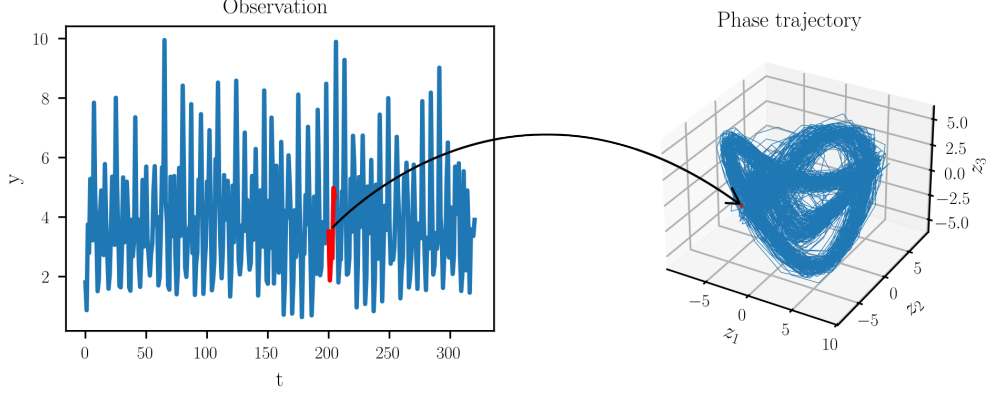


Figure 1 Application of the Takens theorem. On the left are observations $x(t_i)$. To build a point of the phase trajectory $\tilde{\mathbf{x}}(t_i)$ we must take several lagged observation values (colored in red), e.g. $\tilde{\mathbf{x}}(t_i) = (x(t_i), x(t_{i-1}), x(t_{i-2}))^n$.

atmospheric dynamics can also be reduced to an ODEs system [17]. The classification problems in this field include weather regime classification [18] and climate pattern identification [19]. In human activity recognition (HAR) ODEs are primarily used to model the dynamics of human motion. The core idea is that human movement is a continuous physical process governed by laws of mechanics, which can be described by differential equations [20]. This idea can be applied to the basic or fine-grained activity recognition [21, 22], fall detection [23], and other tasks. To conclude, our model is applicable when it is reasonable to assume that observed time series come from some dynamical system of the form (1) or (5).

Our classifier will rely on the knowledge of the vector fields f_j which is not always available. To compensate this we will build approximate fields from data using Neural Ordinary Differential Equations (NODE) [24]. It allows to use gradient optimization when an optimized objective depends on ODE trajectories. There are plenty of papers using NODE for regression tasks. For example, NODE are used in the context of the physics-informed ODE [25, 26] and control learning [27, 28]. However, very few papers analyze NODE in the context of time series classification. Authors of [29] reinterpreted convolutional neural networks in terms of latent ODE dynamics and used

it for hyperspectral image classification. Paper [30] solved classification problem in terms of control theory and NODE. Our paper contributes to the NODE applications for the general dynamical systems.

The NODE paper [24] introduced its own popular time series model called latent ODE. It is illustrated on the Fig. 2. The model is probabilistic and in our terms it can be described as

$$\begin{aligned}\bar{\mathbf{x}}_{t_0} &\sim p_0(\bar{\mathbf{x}}_{t_0}), \\ \bar{\mathbf{x}}_{t_1}, \bar{\mathbf{x}}_{t_2}, \dots, \bar{\mathbf{x}}_{t_N} &= \text{ODESolve}(\bar{\mathbf{x}}_{t_0}, f, t_0, \dots, t_N), \\ \text{each } \mathbf{x}_{t_i} &\sim p_x(\mathbf{x}_{t_i} | \bar{\mathbf{x}}_{t_i})\end{aligned}$$

where $\text{ODESolve}()$ is an approximate solution of the initial value problem using ODE solver [?]. Authors used a variational autoencoder (VAE) [31] to inference the distributions p_0, p_x and the vector field f from the $\mathbf{x}_{t_i}$. While this model is more general and is widely used [32–34], it has several drawbacks compared to ours. First, it has to parameterize f, p_x and the VAE. Hence, more data is needed in general to obtain good approximations of these functions. Our approach may only need to parameterize f . Second, in the latent ODE model a connection between the phase trajectories and the observed time series is given by the general distribution p_x . Thus, the connection is difficult to interpret. In addition, the VAE is needed to reconstruct the $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$. Our reconstruction approaches discussed above give explicit and easy ways to obtain $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$ and vice versa. Actually, several papers can be found in the literature [35, 36] that use Takens theorem ideas with the NODE to restore latent dynamics. But unfortunately, it is poorly covered for machine learning applications. Our paper aims to close this gap.

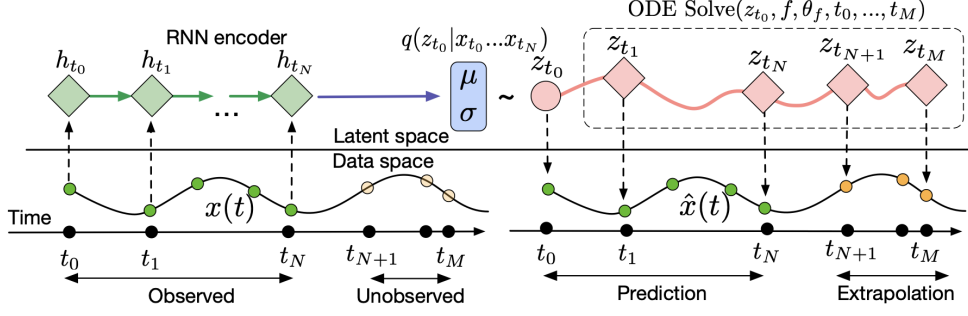


Figure 2 Computational graph of the latent ODE model. Source: reprinted from Chen et al.(2018). Here $z(t_i)$ correspond to the $\bar{\mathbf{x}}(t_i)$.

In the next section we will describe a general principal for the time series classification inside our model. We will build an asymptotically optimal classifier under mentioned properties of the noise ϵ_{t_i} and several additional conditions. These conditions will help with a theoretical prove but may not hold in practice. So we will progressively discard this conditions and adapt classifier for a more general settings. We will also show how adapt the classifier to the SDE case (5). Finally, we will describe the procedure of building vector field approximations from the data. Then we will apply our method to three classification tasks for emprirical validation. The first will be the classification of several known chaotic [?] dynamical systems. The second will be classification of the human motions from the Inertial Measurement Unit (IMU) sensors. The last one will be classification of heart condition from the ECG recordings.

2 Asymptotically optimal classifier

The main principle for building a classifier in the paper is the proximity of an observed time series $\mathbf{x}(t_i)$ to the theoretical initial value problem solution $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)$ for each class $j \in 1 \dots K$. Basically, our classifier is distance-based.

Denote by $\rho_n(\mathbf{x}(t_i), \mathbf{y}(t_i))$ the metric between time series $\mathbf{x}(t_i)$, $\mathbf{y}(t_i)$ of length n defined as

$$\rho_n(\mathbf{x}(t_i), \mathbf{y}(t_i)) = \sum_{i=1}^n \|\mathbf{x}(t_i) - \mathbf{y}(t_i)\|_2^2. \quad (7)$$

Then, define the classifier as

$$\delta(\mathbf{x}(t_i)) = \arg \min_{j \in 1 \dots K} \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)). \quad (8)$$

It turns out that this theoretical classifier has vanishing probability of misclassification as the length of the time series increases. But starting point $\bar{\mathbf{x}}_0$ must be known exactly and phase trajectories from different systems must diverge in some sense. The formal result:

Theorem 1. *Let $\mathbf{x}(t_i) = \{\mathbf{x}_{t_1}, \dots, \mathbf{x}_{t_n}\}$ be the time series to classify defined for all $n \in \mathbb{N}$. Let $\bar{\mathbf{x}}_0$ be the starting point of the associated phase trajectory $\bar{\mathbf{x}}(t_i)$. And let*

$$\rho_\infty(\bar{\mathbf{x}}(t | \bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t | \bar{\mathbf{x}}_0, f_p)) = +\infty \quad \forall j \neq p; \quad j, p \in 1 \dots K.$$

Denote by y the true class of the time series. Then, under proposed conditions on the noise (4), the following holds

$$\mathbb{P}(\delta(\mathbf{x}(t_i)) \neq y) \rightarrow 0, \quad n \rightarrow \infty.$$

The proof of the theorem is deferred to Appendix A.

3 Why optimal classifier may be intractable

Unfortunately, the classifier (8) relies on precise IVP solutions and Theorem 1 relies on precise starting point $\bar{\mathbf{x}}_0$. These requirements are often too strong on practice. We now provide several reasons for that and propose a new classifier.

3.1 Approximate vector fields

Imagine instead of the true vector fields f_j we only have approximate versions $\tilde{f}_j$ such that

$$\|f_j - \tilde{f}_j\| \leq r_j,$$

where $f_j, \tilde{f}_j \in \mathcal{F}(M)$ and r_j are approximation error bounds. This situation always arises when one approximates vector fields from the data (see Section 6).

In general, any small approximation error is enough for phase trajectories from f and $\tilde{f}$ to diverge. We show an Example 1 of such system in Appendix A. The visualization of the associate trajectories is on the Fig. 3 That makes classifier (8) useless. Nonetheless, for a large family of vector fields the $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}_j)$ are arbitrary close to $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)$ if r_j is sufficiently small and times are bounded $t < T$. This is called Grönwall-type bounds and can be found in ODE literature [37]. So further we assume approximation errors are small enough and vector fields f_j are considerably different to not differentiate between $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}_j)$ and $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)$. We also assume that vector fields f_j are considerably different across classes to exclude situations like in Example 1.

3.2 Approximate IVP solutions and noisy phase trajectories

Even if vector fields are known exactly, the IVP is almost always intractable [?]. The numerical methods called ODE solvers are able to approximate IVP solution with arbitrary precision [?] for regular vector fields. The main approximation parameter of the solvers is an integration step τ . A solver has order of convergence l if

$$\sup_{t \in [0, T]} |\text{ODESolve}(t|\bar{\mathbf{x}}_{t_0}, f) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)| = \mathcal{O}(T\tau^l). \quad (9)$$

The higher the convergence order and the smaller τ the higher computational complexity of the ODESolve procedure. Besides this tradeoff some vector fields imply

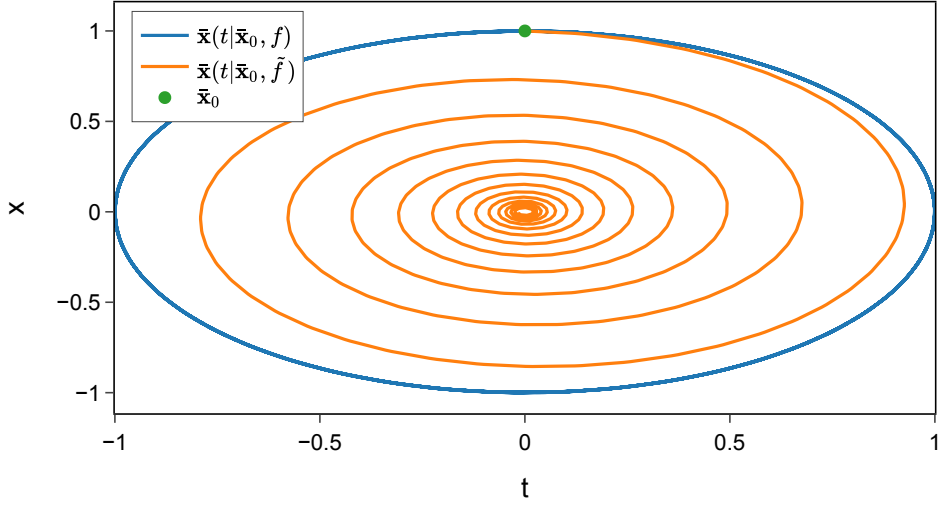


Figure 3 Diverging phase trajectories of the close vector fields $\|f - \tilde{f}\| \leq 0.1$ with $\bar{x}_0 = (0, 1)^T$. While $\bar{x}(t|\bar{x}_0, f)$ is a circle trajectory (called center in dynamical systems), the $\bar{x}(t|\bar{x}_0, \tilde{f})$ is a spiral sink that diverges to the $\mathbf{0}$.

additional constraints on the smallness of the τ . If not satisfied, solver solution diverges from the true one. This effect is called *stiffness* and is usually addressed by choosing an appropriate solver [?]. The illustration of this effect is on the Fig. 4.

As follows from (9) applying ODEsolve to find full trajectory $t \in [0, T]$ starting from the exact $\bar{x}_0$ becomes impractical for long trajectories. We also assume

$$\bar{x}_0 \sim F(\cdot) \quad (10)$$

to be a random vector in general where $F(\cdot)$ is its distribution function. Hence, the classifier 8 can not be applied in these conditions.

On the other hand, imagine we knew $\bar{x}_0$ and $\bar{x}_{t_i}$ for some $i \in 1 \dots n$. We could then approximate $\bar{x}(t|\bar{x}_0, f), t \in [0, t_i)$ using $\text{ODEsolve}(t|\bar{x}_{t_0}, f)$, and approximate $\bar{x}(t|\bar{x}_0, f), t \in [t_i, T]$ using $\text{ODEsolve}(t|\bar{x}_{t_i}, f)$. The solver's accuracy increases. To obtain practical solution, we could simply replace $\bar{x}_{t_i}$ with $\mathbf{x}_{t_i}$. But if the σ from

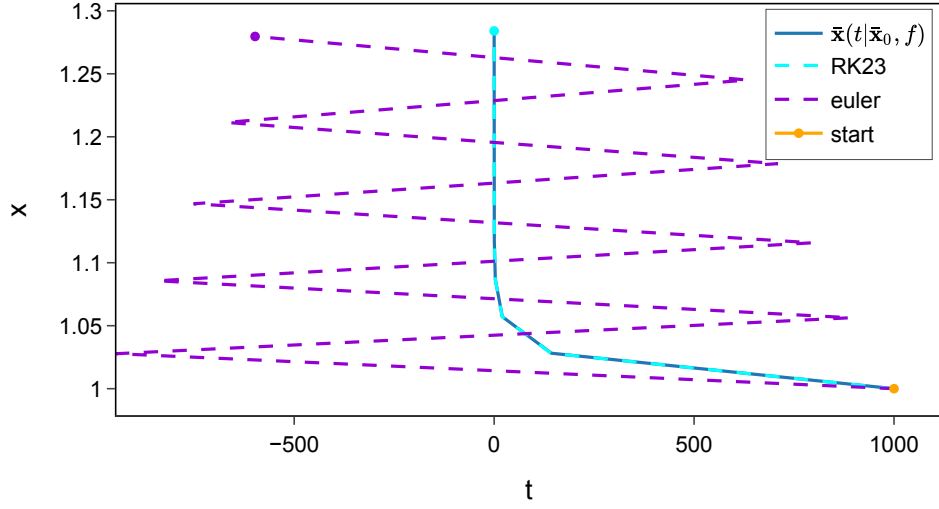


Figure 4 Exact and approximate IVP solutions for the stiff system $f(\bar{\mathbf{x}}) = \begin{pmatrix} -35 & 0 \\ 0 & 0.5 \end{pmatrix} \bar{\mathbf{x}}$ and $\bar{\mathbf{x}}_0 = (1000, 1)^T$. The Runge-Kutta23 (RK23) ODE solver approximates exact solution with high precision. The Euler (euler) ODE solver is unstable and diverges from the exact solution. Integration step $\tau = 0.05$ is equal for both solvers but for Euler it is not small enough. The RK23 has higher convergence order and is stable under such τ .

(4) is large enough then the starting points differ considerably with high probability. This leads to very different IVP solutions $\text{ODESolve}(t|\mathbf{x}_{t_i}, f)$, $\text{ODESolve}(t|\bar{\mathbf{x}}_{t_i}, f)$ in general [?]. We must take randomness into account in a different way.

4 Classification as smoothing problem

Let's break dynamics (1) into consecutive time segments $[t_0, t_1)$, $[t_1, t_2)$, $\dots$, $[t_{n-1}, t_n]$ so we have n IVPs

$$\begin{cases} \frac{d}{dt}\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}}), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_{t_i}, \\ t \in [0, t_{i+1} - t_i]. \end{cases} \quad (11)$$

Let's approximate each IVP solution by the ODEsolve so $\bar{\mathbf{x}}_{t_{i+1}} = \text{ODEsolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f)$. Combining with the (3) and (10) we obtain discrete dynamics

$$\begin{cases} \bar{\mathbf{x}}_{t_{i+1}} = \text{ODEsolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f), \\ \mathbf{x}_{t_i} = \bar{\mathbf{x}}_{t_i} + \epsilon_{t_i}, \\ \bar{\mathbf{x}}_0 \sim F(\cdot). \end{cases} \quad (12)$$

The $\mathbf{x}(t_i)$ is observed, everything else is not. To adapt classifier 8 for this model we need to substitute intractable $\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f)$ for the best approximation based on random $\bar{\mathbf{x}}_0$ and $\mathbf{x}_{t_i}, i \in 1 \dots n$. If we take an L_2 norm in the space of random vectors, the best approximation is *conditional expectation* [?]. We denoted a series of conditional expectations by $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ that means i -th series value is $\mathbb{E}[\bar{\mathbf{x}}_{t_i} | \bar{\mathbf{x}}_0, \mathbf{x}_{t_1}, \dots, \mathbf{x}_{t_n}]$. The problem of finding $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ is called the *smoothing problem*. On the one hand, it is intractable in general [?] similarly to the IVP. On the other hand, there are numerical methods for obtaining approximate solutions. Some of them are extensions of the Kalman filter [?]. Another are based on the bayessian [?] and particle [?] filters. Solving the smoothing problem is an active research area [?].

The new classifier is defined as

$$\delta(\mathbf{x}(t_i)) = \arg \min_{j \in 1 \dots K} \rho_n(\mathbf{x}(t_i), \mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f_j]). \quad (13)$$

The classifier now relies not only on the dynamics driven by the vector fields f_j but also on the observations $\mathbf{x}(t_i)$. This mitigates problems discussed in the Section 3:

1. Inaccuracy from the approximate $\tilde{f}$ and ODEsolve decreases because we give up the full IVP $t \in [0, T]$ in favor of the smaller IVPs on time segments $t \in [t_{i+1}, t_i]$.

2. Randomness of the $\bar{\mathbf{x}}_0, \mathbf{x}_{t_i}$ is taken into account via the conditional expectation

$\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}_{t_i}, f]$. Informally, it optimally balances the underlying dynamics (1) with the noisy observations to approximate the IVP solution $\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f)$.

Comparison of the IVP $\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f)$ approximation by $\text{ODESolve}(t_i|\bar{\mathbf{x}}_{t_0}, f)$ and by $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ is illustrated on the Fig. 5.

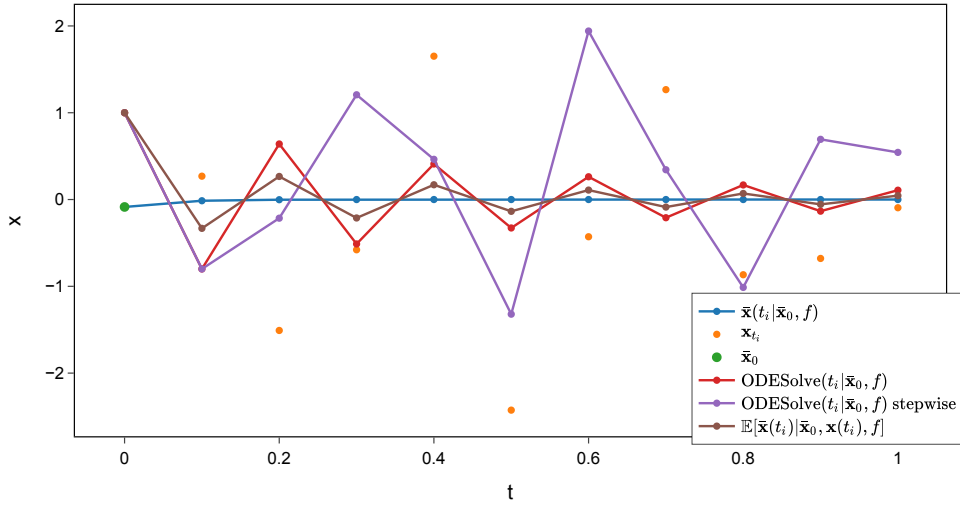


Figure 5 Different IVP approximations for the one-dimensional system $f(x) = -18x$. The initial value $x_0 \sim \mathcal{N}(1, 1)$ and $\epsilon_{t_i} \sim \mathcal{N}(0, 1)$. ODESolve uses Euler solver with initial value $\mathbb{E}[x_0] = 1$. Stepwise ODESolve uses Euler solver on consecutive time segments $[t_i, t_{i+1})$ with noisy initial values x_{t_i} . Unscented Kalman Filter (UKF) [?] is used to compute $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. UKF uses stepwise Euler internally. We can see that UKF demonstrates the best IVP approximation by using both the underlying dynamics and the noisy observations. UKF is the most accurate IVP approximation with probability 0.92 by Monte-Carlo estimation.

5 Adapting classifier for SDE

We can segment dynamics (5) similarly to the (14) and obtain

$$\begin{cases} d\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}})dt + GdW(t), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_{t_i}, \\ t \in [0, t_{i+1} - t_i]. \end{cases} \quad (14)$$

If all time segments are small enough, we can approximate dynamics $f(\bar{\mathbf{x}})dt \approx \text{ODESolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f)$ and $GdW(t) \approx G \cdot \mathcal{N}(0, (t_{i+1} - t_i) \cdot I) = \mathcal{N}(0, (t_{i+1} - t_i) \cdot GG^T) := \xi_{t_{i+1}}$. As a result, we obtain discrete dynamics akin to (12):

$$\begin{cases} \bar{\mathbf{x}}_{t_{i+1}} = \text{ODESolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f) + \xi_{t_{i+1}}, \\ \mathbf{x}_{t_i} = \bar{\mathbf{x}}_{t_i} + \epsilon_{t_i}, \\ \bar{\mathbf{x}}_0 \sim F(\cdot). \end{cases} \quad (15)$$

Finding $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ here is just a more general smoothing problem and can be solved by the same methods listed above. The classifier (13) for the SDE case stays the same. Comparison of the different SDE IVP $\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f)$ approximation is given on the Fig. 6.

6 Learning vector field from data

Sometimes vector field f may be unknown. Instead, time series samples from the associate dynamics (1) may be available. From the (1) and (3) we have

$$\mathbf{x}_{t_i} = \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f) + \epsilon_{t_i} \quad i \in 1 \dots n$$

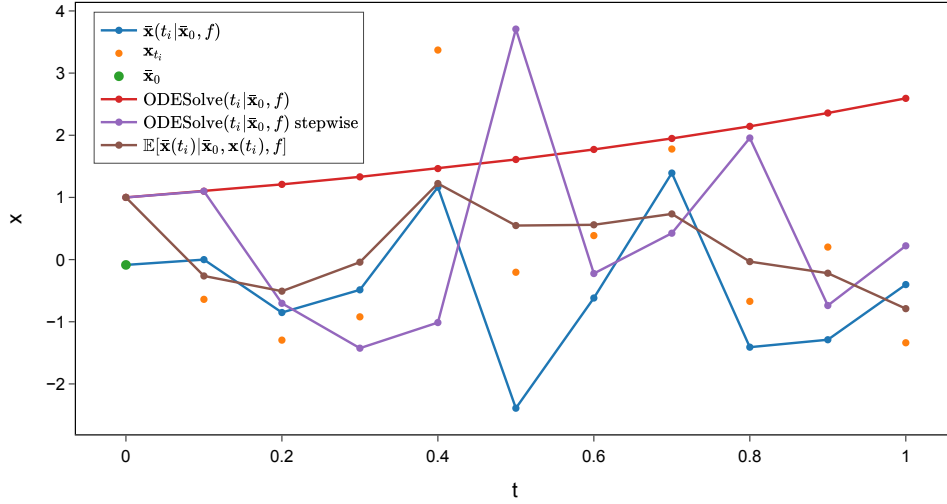


Figure 6 Different IVP approximations for the one-dimensional system $dx(t) = xdt + dW(t)$. The initial value $x_0 \sim \mathcal{N}(1, 1)$ and $\epsilon_{t_i} \sim \mathcal{N}(0, 1)$. Phase trajectory is a random process here in contrast to the ODE dynamics. ODESolve uses Euler solver with initial value $\mathbb{E}[x_0] = 1$. We can see it is inadequate model for the SDE dynamics. Stepwise ODESolve uses Euler solver on consecutive time segments $[t_i, t_{i+1})$ with noisy initial values x_{t_i} . It is more accurate but is subject to the noise and still misses $dW(t)$ term. Unscented Kalman Filter (UKF) is used to compute $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. UKF uses stepwise Euler internally but also account for $dW(t)$ term. It is the most accurate IVP approximation across mentioned with probability 0.97 by Monte-Carlo estimation.

where $\bar{\mathbf{x}}_0$ is defined as in (10). Approximating the true IVP by the ODESolve we obtain

$$\mathbf{x}_{t_i} = \text{ODESolve}(t_i | \bar{\mathbf{x}}_0, f) + \epsilon_{t_i} \quad i \in 1 \dots n.$$

By taking the expectation from both sides and gathering the terms on the left-hand side we get

$$\mathbb{E}[\mathbf{x}_{t_i} - \text{ODESolve}(t_i | \bar{\mathbf{x}}_0, f)] = 0 \quad i \in 1 \dots n.$$

We can apply *generalized method of moments* (GMM) [?] here. Assume we have a set of vector fields $\mathcal{S}_\Theta \subset \mathcal{F}(M)$ parameterized by the real vector Θ . Assume $f \in \mathcal{S}_\Theta$.

Denote the solution of the following optimization problem

$$\sum_{i=1}^n (\mathbf{x}_{t_i} - \text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f))^2 \rightarrow \min_{f \in \mathcal{S}_\Theta} \quad (16)$$

as $f_{\hat{\Theta}}$. Here $\bar{\mathbf{x}}_0, \mathbf{x}_{t_i}$ $i \in 1 \dots n$ are sampled values. If some technical conditions on θ holds [?], the GMM guarantees that

$$f_{\hat{\Theta}} \xrightarrow{\mathbb{P}} f, \quad n \rightarrow \infty. \quad (17)$$

The (16) can be solved using NODE. As discussed in the Section 3, n should not be large due to approximation errors of the ODESolve. Instead, it is better to sample N short batches $B = (\bar{\mathbf{x}}_0, \{\mathbf{x}_{t_i}\}_{i=1}^n)$ and solve

$$\sum_{j=1}^N \sum_{\bar{\mathbf{x}}_0, \mathbf{x}_{t_i} \in B_j} (\mathbf{x}_{t_i} - \text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f))^2 \rightarrow \min_{f \in \mathcal{S}_\Theta} \quad (18)$$

which is equivalent to (16) in terms of the convergence (17).

SDE dynamics (5) can also be learned from the data by the methods similar to NODE [38, 39].

7 Empirical study

In this section we validate the proposed classifier (13) on three classification tasks of progressively decreasing prior knowledge about the underlying dynamics. In the first task the vector fields f_j are known analytically, which lets us isolate the behaviour of the smoothing-based classifier from the error of the field approximation. In the second task the fields are unknown and are learned from data with NODE as described in Section 6. The phase trajectories are still observed directly through a multivariate IMU sensor. Data is taken from the [MotionSense](#) dataset [?]. In the third task the

fields are unknown *and* only a single scalar time series is observed, so the phase trajectories must first be reconstructed via the Takens embedding before learning and classification. The dataset for this task is [ECG5000](#) [?].

7.1 Experimental setup

Across all tasks we evaluate the classifier (13) where the conditional expectation $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f_j]$ is approximated with the Unscented Kalman Filter together with the Rauch–Tung–Striebel smoother [?]. The smoother uses an Euler solver for the ODEsolve in the (12).

Classification quality is reported as accuracy on a held-out test set. All trajectory integration and vector field learning is performed with `torchdiffeq`, the trajectory smoothing with `filterpy`.

7.2 Linear dynamical systems

The first task classifies trajectories generated by three linear ODE systems with the following form

$$f(\bar{\mathbf{x}}) = A\bar{\mathbf{x}} = U\Lambda U^T\bar{\mathbf{x}} = U \begin{pmatrix} -\gamma_1 & w & 0 \\ -w & -\gamma_1 & 0 \\ 0 & 0 & -\gamma_2 \end{pmatrix} U^T\bar{\mathbf{x}}. \quad (19)$$

Here A is a hermitian matrix with specified eigenvalue decomposition. All three systems share the same Λ but differ in eigenvector matrix U . Vector field (19) simulates damped oscillation along first two eigenvectors and exponential decay along the third. We used $\gamma_1 = 0.7, \gamma_2 = 0.6, w = 4.$ For each class we sample $\bar{\mathbf{x}}_0 \sim \mathcal{N}(\mathbf{0}, 4 \cdot I)$ and integrate the system trajectories of length $n = 100$ with step $\tau = 0.05$. Then we corrupt them with additive Gaussian noise of level $\sigma = 2$. following (3)–(4). The visualization

of the final series and underlying phase trajectories is on the Fig. 7 (with lower noise level for clearer illustration).

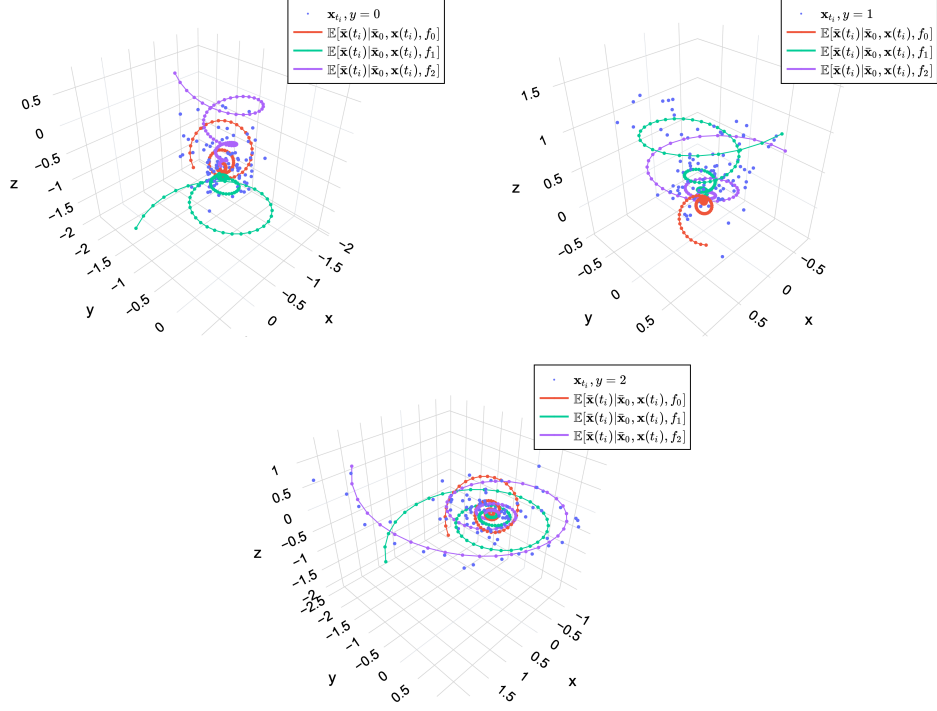


Figure 7 Observed time series $\mathbf{x}(t_i)$ generated from each linear ODE system (y label in the figure legend) and smoothed phase trajectories from every field f_0, f_1, f_2 . The systems are a combination of exponential decay and damped oscillations. Noise level for the illustration is $\sigma = 0.2$.

Because the true fields f_j are available, no field learning is required. We compare our classifier against time series classification baselines: a k -Nearest-Neighbours classifier with the Dynamic Time Warping distance [2] and a recurrent (LSTM) network trained end-to-end on the raw series. Training curves for the LSTM are represented on the Fig. 8. Classification accuracy is reported in the Tab. 1.

From 8 we can see that LSTM was not able to learn ODE systems. The model can't differentiate them due to the high noise level and system's "proximity" implied by the shared Λ . The k -NN + DTW classifier is a distance-based like 13 but is sub-optimal. It uses DTW to compute series distance instead of (7) and $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. It also

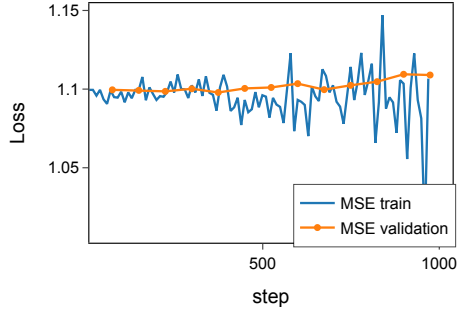


Figure 8 Learning curves for the LSTM training: MSE loss on train and validation series.

Table 1 Classification accuracy on the linear ODE systems task ($K = 3$, $n = 100$, $\sigma = 2$).

Method	Accuracy
k -NN + DTW	0.67
LSTM	0.27
Ours (UKF smoothing)	0.86

learns f_j only indirectly by memorizing training series for each class. Classifier (13) combined information from both f_j and noise level to show the best performance. Though it is not absolutely precise due to the noise level and limited trajectory length.

7.3 Human motion recognition from IMU

The second task uses the MotionSense dataset of Inertial Measurement Unit (IMU) recordings. Here 23 subjects perform 6 activities: walking, jogging, walking upstairs, walking downstairs, standing and sitting. Each activity forms classification class. We take rotation rate (x, y, z) and acceleration (x, y, z) from the IMU sampled at 50 Hz. We assume them to be a noisy phase trajectories like in (3). Unlike the previous experiment, the vector fields for the activities are unknown. So for each class we learn an approximate field $\tilde{f}_j$ from short trajectory batches by solving the objective (18). The noise level σ is estimated directly from the data.

Table 2 Activity classification accuracy on MotionSense IMU data.

Method	Accuracy
k -NN + DTW	<i>TODO</i>
LSTM	<i>TODO</i>
Ours, learned ODE field	<i>TODO</i>
Ours, learned SDE field	<i>TODO</i>

7.4 ECG heart condition classification

The third task classifies heartbeats from the ECG5000 dataset, where each sample is a *single* scalar time series and the class labels distinguish normal beats from several arrhythmia types. Since only one observed coordinate is available, we first reconstruct the phase trajectories with the Takens delay embedding of Section 1 (delay dimension 10, window size 30, step $\tau = 2 \cdot 10^{-3}$), turning the scalar series into delay vectors $(x(t_i), x(t_{i-1}), \dots)$ that live on a diffeomorphic copy of the underlying manifold. We then learn a vector field $\tilde{f}_j$ per class on the reconstructed trajectories via (18) and classify with (13). This is the most demanding setting, combining field approximation with manifold reconstruction.

Table 3 Heartbeat classification accuracy on ECG5000 (Takens-reconstructed dynamics).

Method	Accuracy
k -NN + DTW	<i>TODO</i>
LSTM	<i>TODO</i>
Ours (Takens + learned field)	<i>TODO</i>

7.5 Discussion

8 Conclusion

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Appendix A Proofs and examples

Proof of Theorem 1. First, estimate conditional probability of the misclassification

$$\begin{aligned} \mathbb{P}(\delta(\mathbf{x}(t_i)) = j | y = p) &= \mathbb{P}\left[\bigcap_{s=1, s \neq j}^K \{\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) < \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_s))\}\right] \leq \\ &| \text{monotonicity of probability} | \leq \mathbb{P}[\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) < \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))]. \end{aligned}$$

Expand metrics in the last event using $y = p$:

$$\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p)) = \sum_{i=1}^n \|\mathbf{x}(t_i) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p)\|^2 = \sum_{i=1}^n \|\epsilon_{t_i}\|^2,$$

$$\begin{aligned}
\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) &= \sum_{i=1}^n \|\mathbf{x}(t_i) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)\|^2 = \sum_{i=1}^n \|(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) + \boldsymbol{\epsilon}_{t_i}\|^2 = \\
&= \sum_{i=1}^n \|\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)\|^2 + 2 \sum_{i=1}^n \langle \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle + \sum_{i=1}^n \|\boldsymbol{\epsilon}_{t_i}\|^2.
\end{aligned}$$

Here $\langle \cdot, \cdot \rangle$ is the standard scalar product. The last event is now

$$\begin{aligned}
\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_i)) &< \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) \Leftrightarrow \\
\sum_{i=1}^n \langle \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle &< -\frac{1}{2} \sum_{i=1}^n \|\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)\|^2 \Leftrightarrow \\
\sum_{i=1}^n \langle \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle &< -\frac{1}{2} \rho_n(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p)).
\end{aligned}$$

Apply Markov inequality and use (4):

$$\begin{aligned}
\mathbb{P}[\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) &< \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))] &\leq 4 \frac{\mathbb{D}[\sum_{i=1}^n \langle \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle]}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n \mathbb{E}[\langle \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle^2]}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n (\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j))^T \mathbb{E}[\boldsymbol{\epsilon}_{t_i} \boldsymbol{\epsilon}_{t_i}^T] (\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j))}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n (\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j))^T R (\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j))}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n \langle \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), R (\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)) \rangle}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} \leq \\
&\leq 4\sigma^2 \frac{\sum_{i=1}^n \|\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)\|^2}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} = 4\sigma^2 \frac{\rho_n(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))}{\rho_n^2(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} = \frac{4\sigma^2}{\rho_n(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p))} := \xi_n.
\end{aligned}$$

By theorem condition we have

$$\rho_n(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p)) \rightarrow \rho_\infty(\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_p)) = +\infty, \quad n \rightarrow \infty$$

so $\xi_n \rightarrow 0$, $n \rightarrow \infty$ for all $j, p \in 1 \dots K$.

Second, rewrite probability of the misclassification via conditional probability:

$$\begin{aligned} \mathbb{P}(\delta(\mathbf{x}(t_i)) \neq y) &= \mathbb{E}_y \left[\sum_{p=1}^K \mathbb{P}(\delta(\mathbf{x}(t_i)) \neq p | y = p) \right] = \mathbb{E}_y \left[\sum_{p=1}^K \mathbb{P} \left(\bigcup_{j=1, j \neq p}^K \{\delta(\mathbf{x}(t_i)) = j | y = p\} \right) \right] \leq \\ | \text{union bound} | &\leq \mathbb{E}_y \left[\sum_{p=1}^K \sum_{j=1, j \neq p}^K \mathbb{P}(\delta(\mathbf{x}(t_i)) = j | y = p) \right] \leq \sum_{p=1}^K \sum_{j=1, j \neq p}^K \xi_n = K(K-1)\xi_n \rightarrow 0. \end{aligned}$$

□

Example 1. Here we construct a system of two arbitrary close vector fields with diverging phase trajectories. Let $M = \{\bar{\mathbf{x}} \in \mathbb{R}^2 \mid \|\bar{\mathbf{x}}\|_2 \leq R\}$ be a plane circle. Define the vector field's norm as a sup norm

$$\|f\| = \sup_{\bar{\mathbf{x}} \in M} \|f(\bar{\mathbf{x}})\|_2.$$

Let $\bar{\mathbf{x}}_0 = c \cdot (\cos \gamma, \sin \gamma)^T \neq \mathbf{0}$ be an initial value. Let $\{t_i\}_{i=1}^\infty$ be any time sequence and let $r > 0$ be an arbitrary small value. Construct two vector fields

$$\begin{aligned} f(\bar{\mathbf{x}}) &= \begin{pmatrix} ia & 0 \\ 0 & ia \end{pmatrix} \bar{\mathbf{x}} = A\bar{\mathbf{x}}, \\ \tilde{f}(\bar{\mathbf{x}}) &= \begin{pmatrix} ia - b & 0 \\ 0 & ia - b \end{pmatrix} \bar{\mathbf{x}} = \tilde{A}\bar{\mathbf{x}} \end{aligned}$$

where i is an imaginary unit, $a \neq 0, b > 0$ are some scalars. Then

$$\|f - \tilde{f}\| \leq \|A - \tilde{A}\|_2 \cdot \|x\| = \left\| \begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix} \right\|_2 \cdot \|x\| \leq bR$$

where $\|A - \tilde{A}\|_2$ is matrix spectral norm. So if $b < \frac{r}{R}$ then $\|f - \tilde{f}\| < r$. Next, it easy to obtain that IVP solutions for the ODE systems are

$$\begin{aligned} \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f) &= c \begin{pmatrix} \sin(\gamma - at) \\ \cos(\gamma - at) \end{pmatrix}, \\ \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}) &= c \exp(-bt) \begin{pmatrix} \sin(\gamma - at) \\ \cos(\gamma - at) \end{pmatrix}. \end{aligned}$$

It is obvious that $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}) \rightarrow \mathbf{0}, t \rightarrow \infty$ and $\|\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f})\| = \|\bar{\mathbf{x}}_0\| = c \forall t$. So

$$\|\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})\| \rightarrow c \neq 0, i \rightarrow \infty$$

and then

$$\rho_n(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})) = \sum_{i=1}^n \|\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})\|_2^2 \rightarrow \infty, n \rightarrow \infty$$

by the absence of the necessary condition for series convergence.

As a result, classifier (8) that uses $\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})$ do not have guarantees of the Theorem 1 even though f and $\tilde{f}$ are arbitrary close.